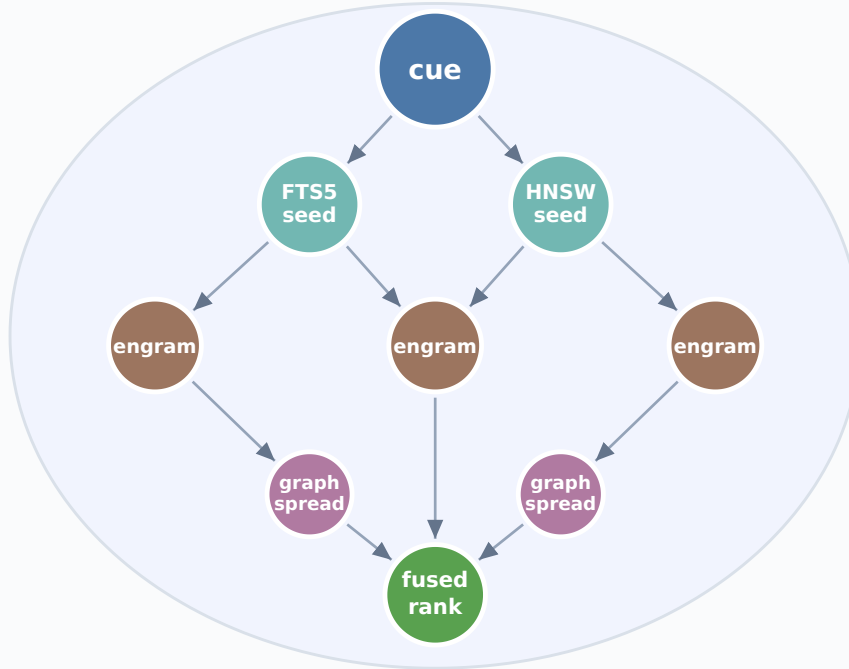


### (a) Cue-driven memory activation



*activate(cue)*

● write path · experience()    ● read path · activate(cue)

### (b) Agent brain directory

```
manifest.json
hippocampus/      engram segments
graph/            synapse edges
recall_index.sqlite
  FTS5 + vector blobs
*.wal              → checkpoint()
```

↓ stores engram uni

### (c) Engram (logical unit)

```
content
.
context
.
outcome
provenance      kind, verified
rag             doc_id, chunk_id
semantic_vector optional embedding
graph wiring     co-activation edges
```

↓ upsert to sidecar

### (d) Recall sidecar

FTS5 lexical seeds + HNSW semantic seeds

### (e) Write: experience()

1. separation gate
2. encode episode
3. wire graph synapses
4. upsert recall sidecar

### (f) Read: activate(cue)

1. lexical + semantic seed
2. graph spread (0–4 hops)
3. fuse FTS5 + vector scores
4. provenance boost → rank

### (g) Third data model

#### Relational

- Typed rows
- Predicate match
- No graph

#### Vector DB

- ANN top-k
- Embedding in
- No provenanc